NCERT • GRADE 7 • SCIENCE

Transport in Plants

C10 • Life Processes in Plants

Roots • Xylem • Phloem

Stomata • Food transport • Experiments

Chapter 10 transport companion notes

Textbook solutions first • Visual understanding • Revision

Original student notes based on NCERT. Illustrations are editable LaTeX drawings.

Part A • NCERT Chapter 10 textbook questions (pp. 150–152)

How these notes fit the collection

These notes complete the **transport and respiration in plants** portions of Chapter 10. Your existing `C10_Photosynthesis.tex` explains photosynthesis in depth. All ten textbook exercise questions are answered here as well, with extra attention to **Q3 and Q9**, which concern transport.

Q1 • Complete the photosynthesis–respiration table

Feature	Photosynthesis	Respiration
Raw materials	Carbon dioxide and water	Glucose and oxygen
Products	Glucose and oxygen	Carbon dioxide, water and energy
Word equation	CO_2 + water → glucose + O_2	Glucose + oxygen → CO_2 + water + energy
Importance	Makes food, stores light energy	Releases energy usable for living processes

Photosynthesis requires sunlight and chlorophyll. These are overall word equations, not balanced chemical equations.

Q2 • What if photosynthetic organisms disappear?

The food supply at the base of most food chains would collapse. Oxygen production would decline while carbon dioxide could rise, damaging ecosystems and the survival of many organisms. This would not mean all oxygen vanished immediately.

Q3 • Why does a potato contain starch?

Food is made mainly in **green leaves** during photosynthesis. Sugars are transported through **phloem** from leaves to developing potato tubers, where they are converted into **stored starch**. The **xylem** does not carry this prepared food; it carries water and minerals.

Q4 • Why are broad, flat leaves helpful?

A large flat surface generally receives more sunlight, allowing more chlorophyll-containing cells to absorb light. Stomata permit gas exchange. Actual photosynthesis also depends on water, temperature and carbon dioxide.

Q5 • Identify X, Y and Z

X = glucose, Y = oxygen, Z = water. Thus: glucose + oxygen → carbon dioxide + water + energy. This represents respiration, not photosynthesis.

Q6 • Two plants in sunshine and darkness

(i) Krishna tests the need for **sunlight** to make food by photosynthesis. (ii) Over time the shaded plant may lose vigour; the change is not necessarily immediate. (iii) After first **destarching both plants** and providing the test conditions, a leaf from the sunlit plant turns blue-black with iodine, showing newly formed starch. The dark-kept leaf does not show newly formed starch.

Q7 • Light and carbon dioxide: four setups

(i) Newly formed starch: **(a)** sunlight with CO_2. (ii) No newly formed starch: **(b), (c), (d)**. (iii) Photosynthetic oxygen generation: **(a)**. (iv) No photosynthetic oxygen generation: **(b), (c), (d)**. Other cellular respiration can still occur in every living plant.

Q8 • Water plant, snail and indicator

Ananya is examining how **respiration and photosynthesis affect carbon dioxide in water**. The snail releases CO_2; the plant may use it during photosynthesis in sufficient light, but also respires. Compare A (snail), B (plant), C (both) and D (water control). Record the indicator's *actual* colour changes and compare them with the supplied colour chart; the colours cannot be predicted without knowing which indicator is used. Keep light, water volume and time comparable.

Q9 • Warm versus cold: design a fair test

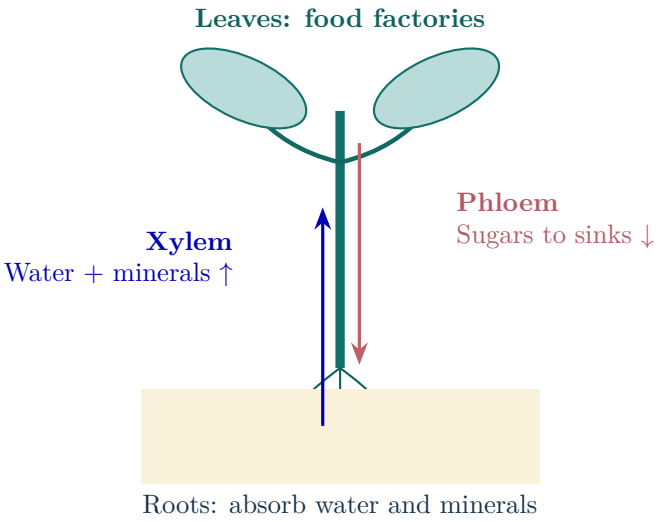
Question: Does water reach a plant stem/leaf faster at a warmer temperature?
Method: (1) Take two similar freshly cut twigs from one species, preferably with white flowers. (2) Place the cut ends in equal volumes of water coloured with the *same* amount of red food colouring. (3) Keep one at a *moderately warm* temperature and one at a cooler temperature; use similar light and airflow. (4) Mark equal heights and record when colour becomes visible there. Repeat several trials.
Measurements: Compare elapsed times to reach the same height, or heights reached in the same time. Smaller time means quicker transport.
Control variables: twig size/species, water level, dye concentration, light, airflow and observation interval.
Conclusion: Record the evidence first. Warming may increase transpiration and water movement, but the result must be measured, not assumed; extreme heat can harm the plant. **Safety:** do not use hot/boiling water or damage living plants.

Q10 • How do photosynthesis and respiration maintain balance?

Photosynthesis uses carbon dioxide and releases oxygen while making food. Respiration uses oxygen to break down food, releasing carbon dioxide, water and energy. Together these processes help cycle materials and transfer energy through living systems. **Plants perform respiration both day and night.**

Part B • Transport in plants: the big idea

Plants have two different transport systems. Roots absorb water and dissolved minerals. Leaves make sugars through photosynthesis. These substances must reach other parts of the plant.



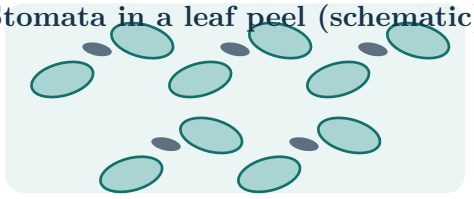
Do not mix up the two tubes

Xylem: roots → stems and leaves; mainly upward movement of water and minerals.
Phloem: carries sugars/food from where they are produced or stored (sources) to where they are needed (sinks); it is *not restricted to downward* transport.

Part C • NCERT Activities 10.6 and 10.7

Activity 10.6 • Looking for stomata (teacher demonstration)

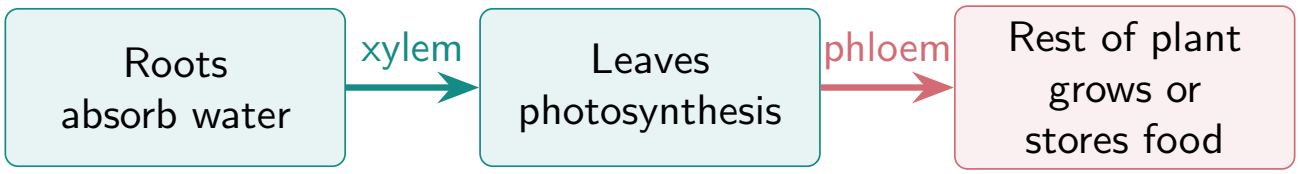
A thin peel from the lower leaf surface is mounted in water and examined under a microscope. **Observation:** tiny openings (stomata) are seen. **Inference:** pores help gases enter and leave leaves. Stomata also allow water vapour to escape. Do not confuse stomata (pores) with xylem/phloem (transport tissues).



Activity 10.7 • Red dye moving through a twig

Setup: keep two similar twigs in water; add red colouring only to tumbler B. Leave them for a day. **Observation:** red colour can appear in the stem, leaf veins or pale flowers of twig B. Cut the stem and observe coloured strands.
Inference: dissolved colour travels with water through **xylem**. This experiment does not directly prove the movement of sugars through phloem.

Part D • Photosynthesis, respiration and transport



Respiration: all living plant parts release energy from food, day and night

Part E • Common mistakes and quick revision

Common mistakes → corrections

- "Xylem carries food." → **Xylem carries water and minerals.**
- "Phloem only moves food down." → Food can move in different directions from sources to sinks.
- "Plants breathe only at night." → **Respiration occurs day and night.**
- "A potato makes its starch without leaves." → Potato stores carbohydrates transported from photosynthetic parts.

Part F • Student scientist: record real evidence

Compare transport fairly: observation sheet

Use this small table during the temperature experiment from Q9. Write an *observation*, not a guessed result.

Condition	Trial 1	Trial 2	Trial 3
Cool twig: time to dye mark	_____ s	_____ s	_____ s
Warm twig: time to dye mark	_____ s	_____ s	_____ s

Three scientific-thinking checks

Why is clear water needed in the control? So any red colour can be connected with added dye.
Does coloured water travel in phloem? The dye evidence in this activity points to **xylem** transport.
Why do several trials? Repeated measurements help reveal inconsistent results and reduce chance effects.

Four exam-ready practice questions (additional, not NCERT)

1. Name the tissue moving water upwards. **Xylem.**
2. Name the tissue transporting food. **Phloem.**
3. Which organ absorbs water from soil? **Roots.**
4. Why can roots grow underground? **Phloem brings them food.**

Reference: *NCERT Curiosity*, Grade 7, Chapter 10, pp. 147–152.